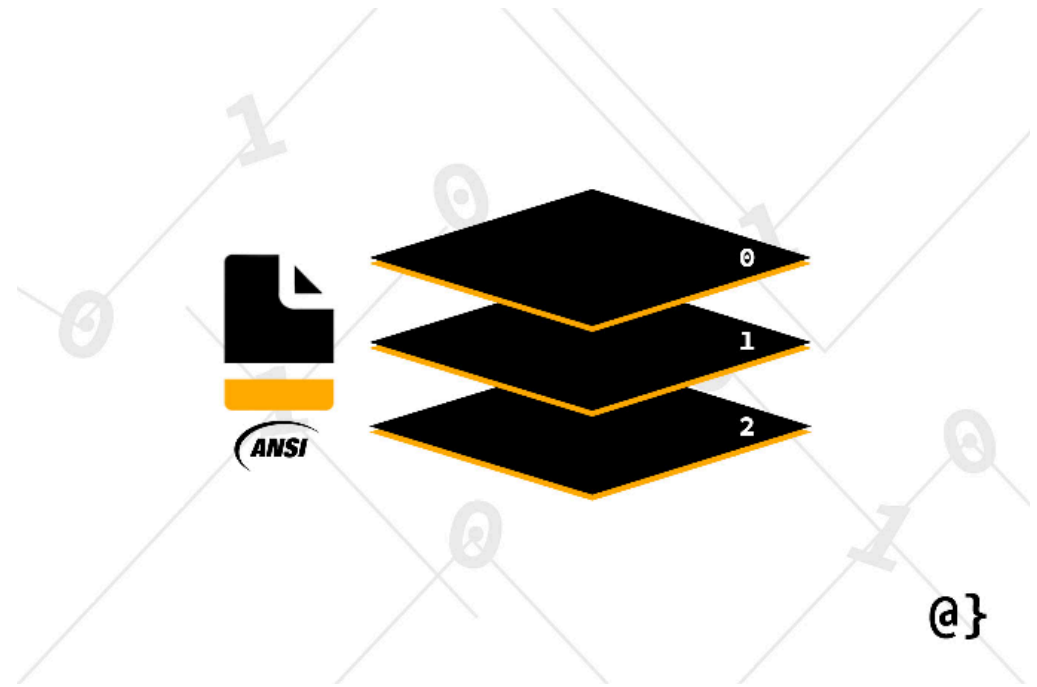


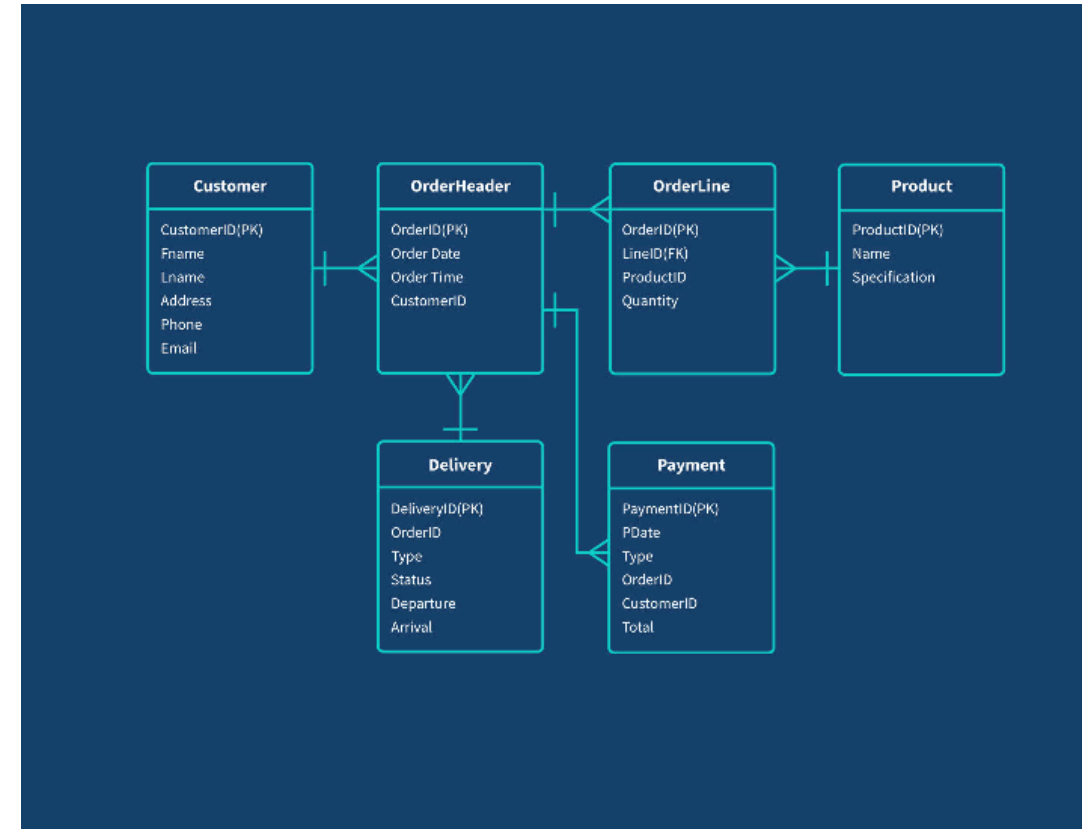
Different Data Models: Hierarchical, Network, and Relational

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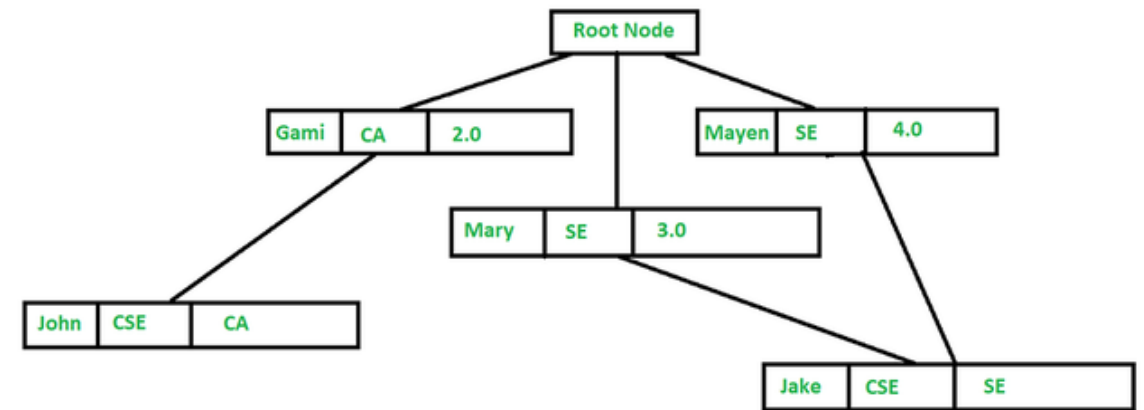
Introduction

- A **data model** defines how data is stored and organized.
- It helps users manage data efficiently.
- The three common data models are:
- Hierarchical Model
- Network Model
- Relational Model



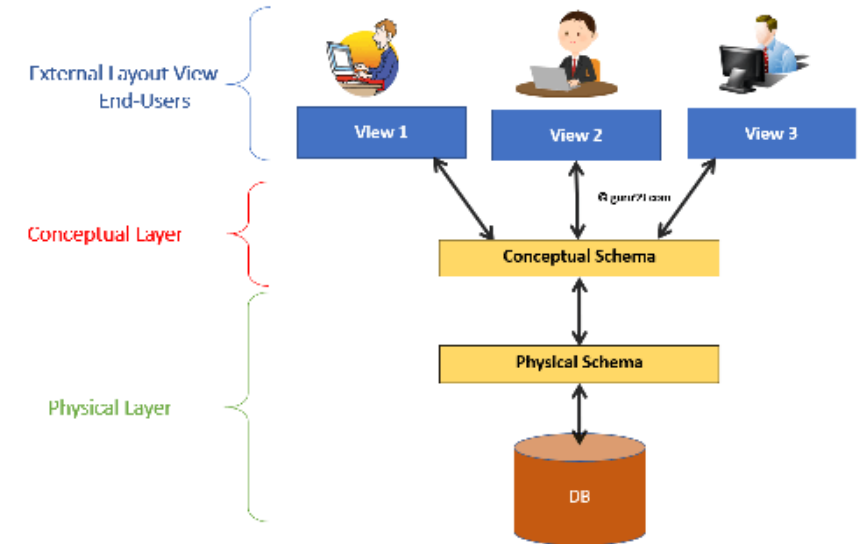
Hierarchical Data Model

- Data is organized in a **tree structure**.
- It has **Parent** and **Child** relationships.
- Each child has only **one parent**.
- Easy to understand.



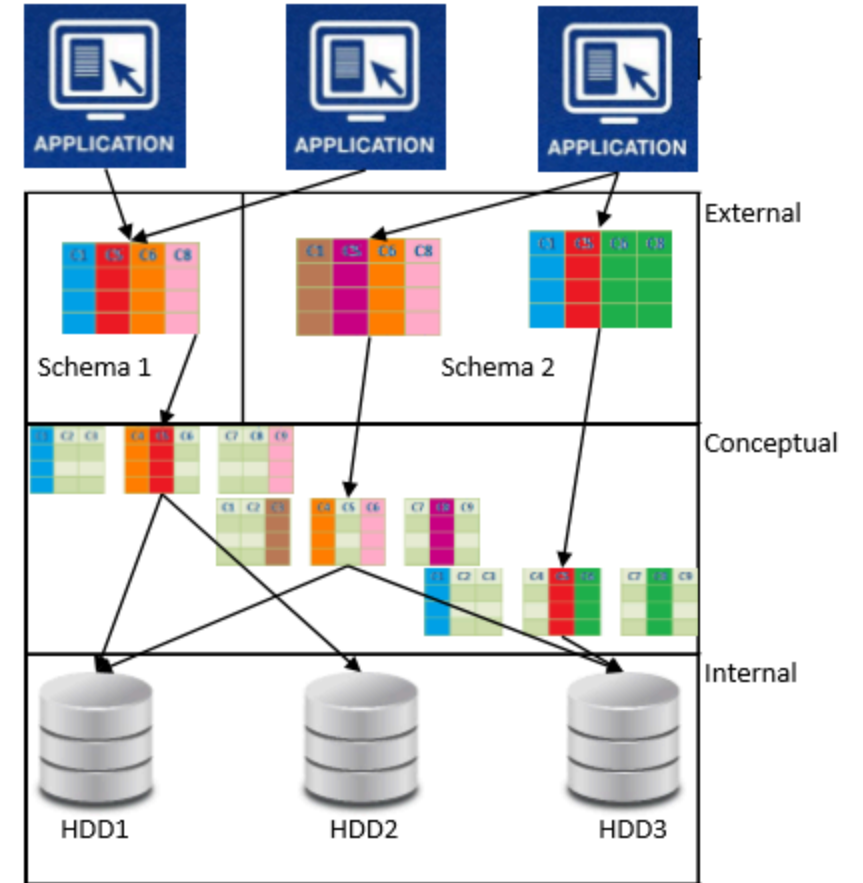
Network Data Model

- Data is connected like a **network**.
- A **Teacher** can have **multiple Students** .
- Supports many-to-many relationships.
- More flexible than the hierarchical model.



Relational Data Model

- Data is stored in **tables**.
- Tables contain **Rows** and **Columns**.
- Tables are connected using **Primary Key** and **Foreign Key**.
- Most widely used database model.



Comparison of Data Models

<u>Feature</u>	<u>Hierarchical</u>	<u>Network</u>	<u>Relational</u>
Structure	Tree	Graph	Tables
Relationship	One-to-Many	Many-to-Many	Table Relations
Complexity	Low	High	Medium
Flexibility	Low	High	High
Ease of Use	Easy	Difficult	Easy

Real-World Applications

- Data Model

- Hierarchical
- Hierarchical
- Network
- Network
- Relational
- Relational
- Relational

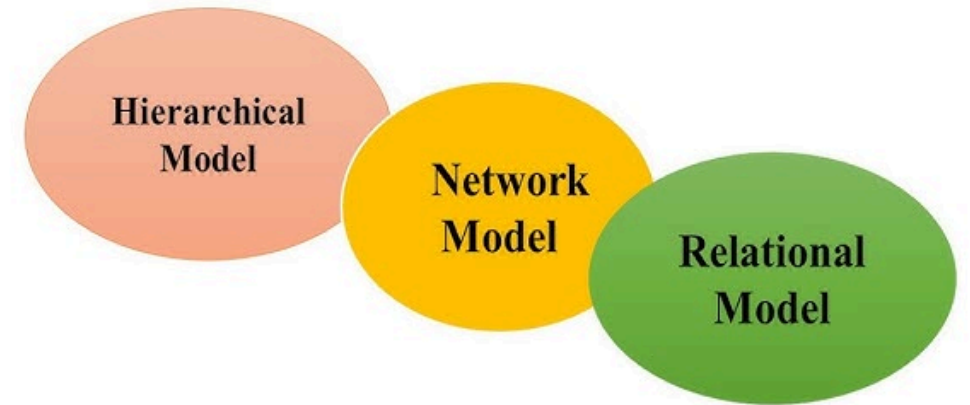
- Example Industry

- Organization Structure
- Windows File System
- Airline Reservation
- Banking Network
- Amazon
- Hospital Management
- College Database

Conclusion

- Data models organize information efficiently.
- Hierarchical model uses a tree structure.
- Network model allows multiple relationships.
- Relational model stores data in tables and is the most widely used.
- Choosing the correct data model improves database performance and management.

Database Models



Thank
you!